Used Vehicle Search

Technische Universität München

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Python and Advanced Data Science

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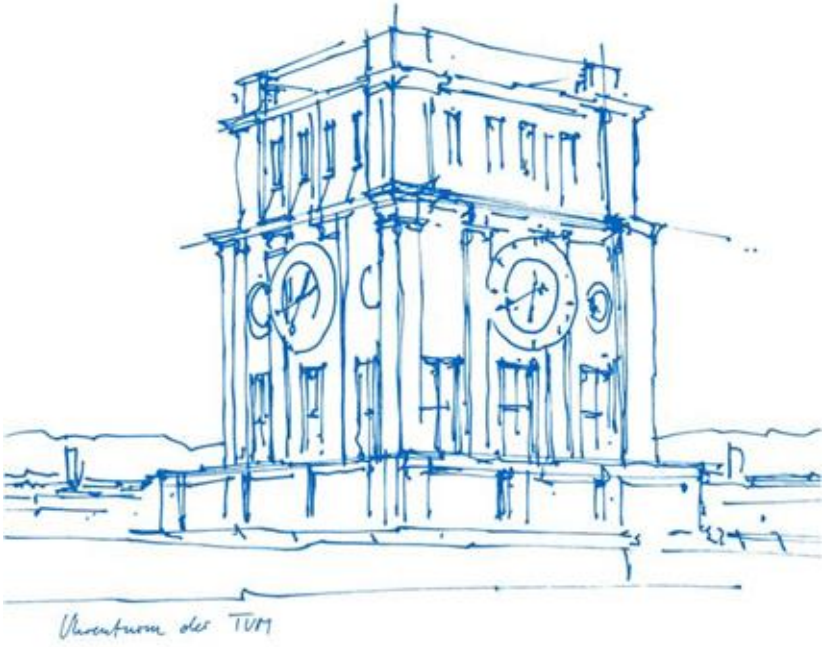


Table of Content

1. Background
2. Data Preparation
3. Explorative Data Analysis
4. Model Training and Evaluation
5. Limitations and Further Development
6. Recommendations

1. Background

- Introduction & Motivation
- Framework of the Project

1.1 Introduction & Motivation

📊**Our Dataset: Used Vehicles**

The dataset was created by scraping used vehicle listings on Craigslist across the United States. It includes detailed information on car sales, such as price, condition, manufacturer, and 20 other categories. The data is updated every few months, providing a comprehensive resource for analysis.

📈 **Market Challenges (Motivation):**

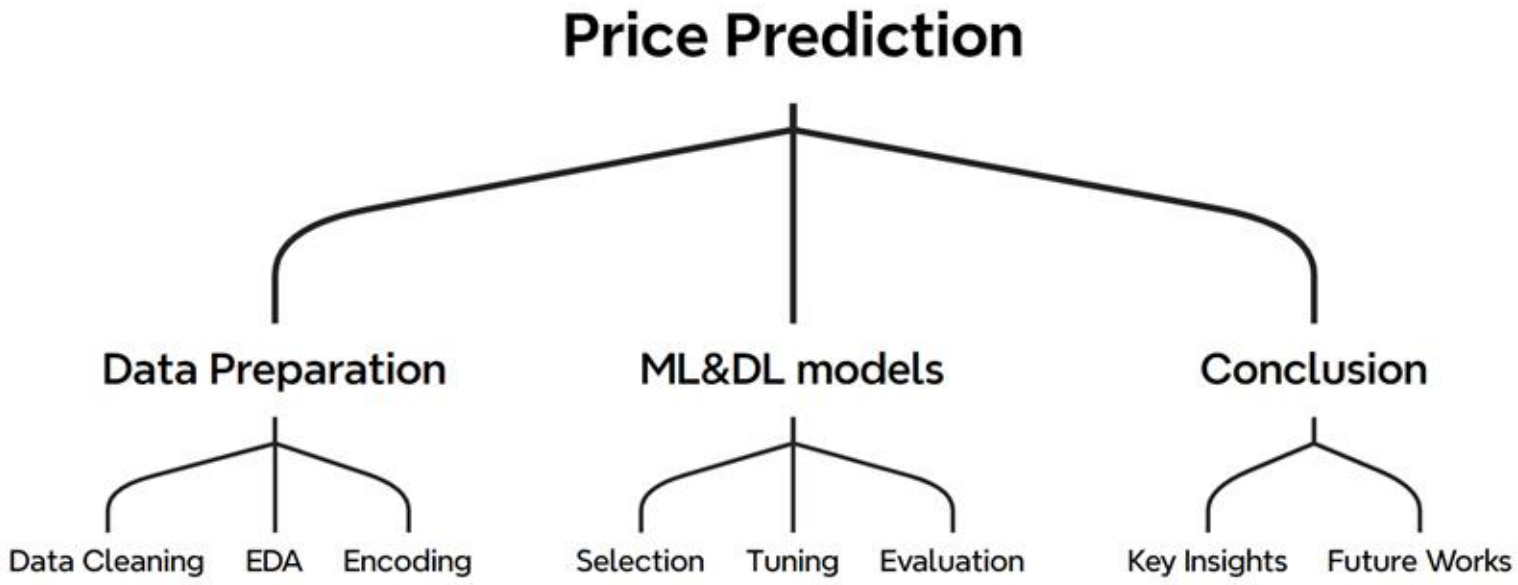
- Pricing influenced by multiple factors *(mileage, condition, demand……)*.
- Sellers lack reliable pricing guidance.

🎯 **Goal:**

G1: Provide actionable insights for most related factors.

G2: Develop a data-driven model to enhance price prediction.

1.2 Framework of the Project



2. Data Preparation

The goal of this section is to handle missing values and outliers, preparing the data for subsequent processing. Based on our knowledge of the used car and the data distribution, we applied scientifically-driven methods to handle the original dataset.

2.1 Overview

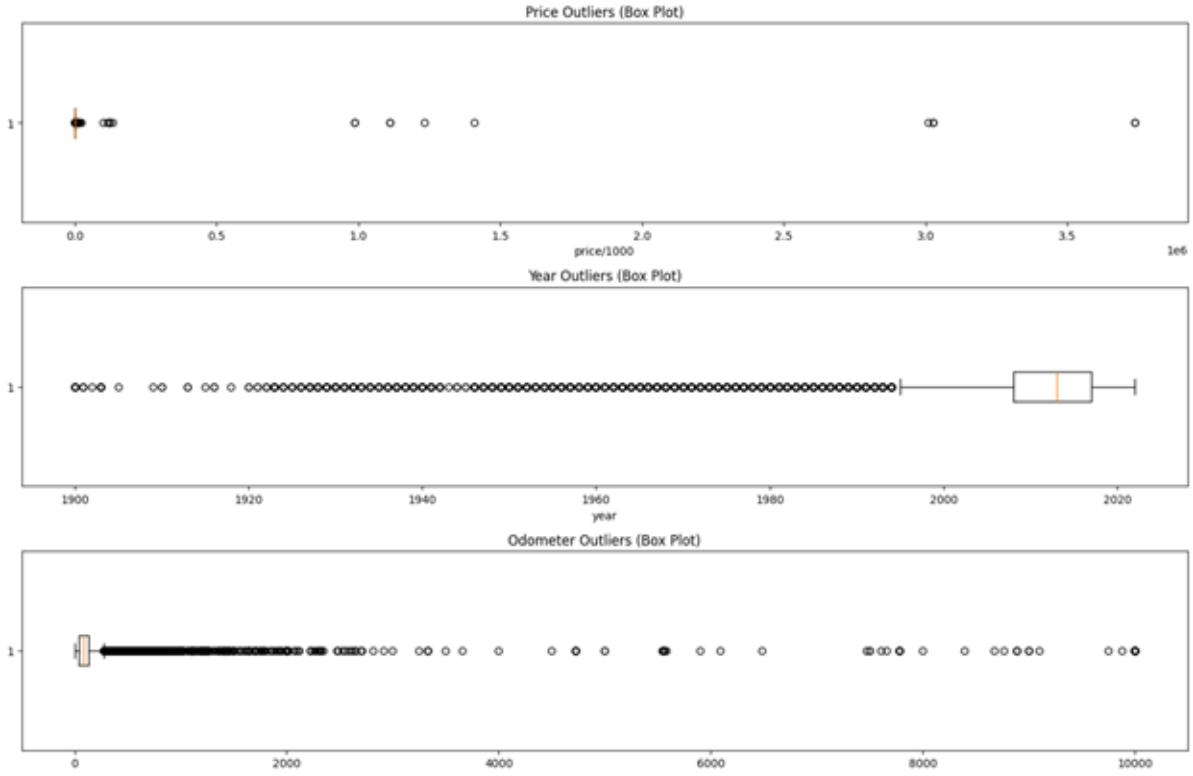
```
Data columns (total 26 columns):
 #   Column        Non-Null Count   Dtype
---  ------        --------------   -----
 0   id            426880 non-null  int64
 1   url           426880 non-null  object
 2   region        426880 non-null  object
 3   region_url    426880 non-null  object
 4   price         426880 non-null  int64
 5   year          425675 non-null  float64
 6   manufacturer  409234 non-null  object
 7   model         421603 non-null  object
 8   condition     252776 non-null  object
 9   cylinders     249202 non-null  object
 10  fuel          423867 non-null  object
 11  odometer      422480 non-null  float64
 12  title_status  418638 non-null  object
 13  transmission  424324 non-null  object
 14  VIN           265838 non-null  object
 15  drive         296313 non-null  object
 16  size          120519 non-null  object
 17  type          334022 non-null  object
 18  paint_color   296677 non-null  object
 19  image_url     426812 non-null  object
 20  description   426810 non-null  object
 21  county        0 non-null       float64
 22  state         426880 non-null  object
 23  lat           420331 non-null  float64
 24  long          420331 non-null  float64
 25  posting_date  426812 non-null  object
```

What needs to be done?

- Data type
- Missing values
- Outliers
- Unwanted variables

2.2 Data Cleaning

Filter out outliers using both statistics and practical knowledge



price: adjusted IQR

year>1920
vehicle_age: [0,25)

odometer:
(100,1000000]

2.2 Data Cleaning

Apply different methods to handle missing values

Variable	Missing Percentage	Method Applied
fuel	0.71%	drop the missing rows
odometer	1.03%	fill with median
title_status	1.93%	fill with mode ‘clean’
manufacturer	4.13%	drop the missing rows
condition	40.79%	KNN
cylinders	41.62%	fill according to model
size	71.77%	remove the variable

3. Explorative Data Analysis

In Exploratory Data Analysis (EDA), we aim to identify key factors influencing used car prices and examine the relationships between independent variables. Additionally, we will encode categorical variables and finalize the dataset to ensure it is ready for modeling.

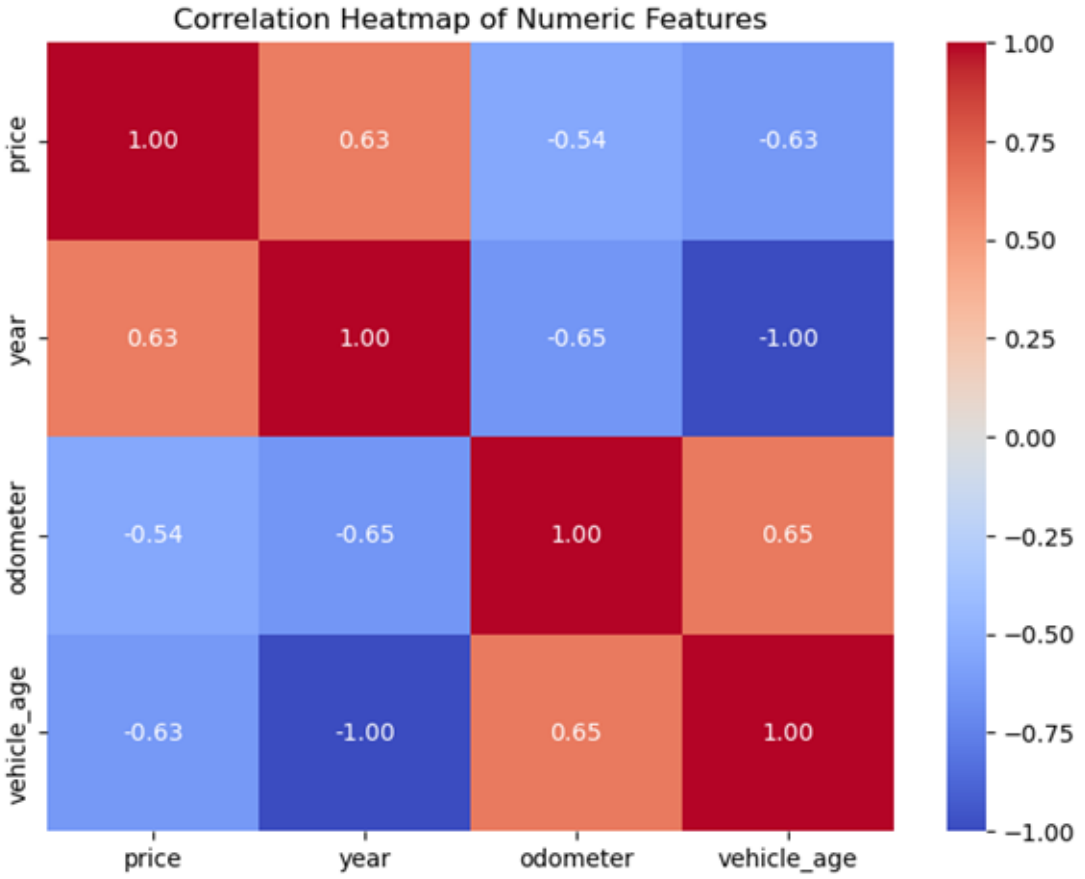
3.1 Correlation Analysis

Object:
To avoid redundancy and improve model stability, we analyzed collinearity among numerical variables.

Result Explanation:
The result shows that odometer and price have a negative correlation of -0.54, indicating that higher mileage is generally associated with lower prices.

Vehicle age and price exhibit a negative correlation of -0.63, meaning older cars tend to be less expensive.

Vehicle age and year show strong collinearity.

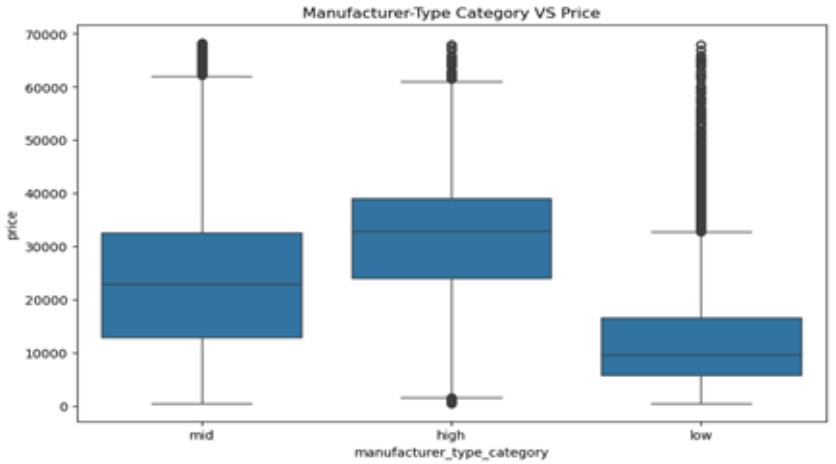


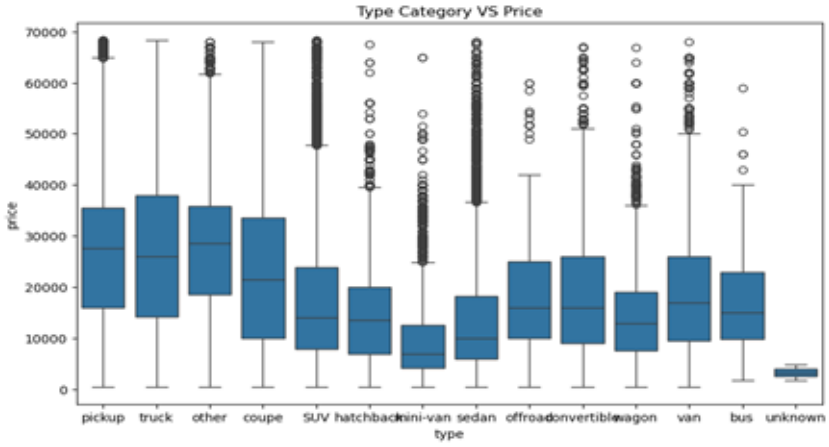
3.2 Categorical Variables and Price Analysis

Result Explanation:

I initially categorized manufacturers into low, mid, and high-end groups based on common knowledge. However, the box plot revealed that some low-end brands had even higher prices than mid-tier ones, suggesting additional factors at play.

After I combined manufacturer with type, the box plot showed a more expected price distribution.





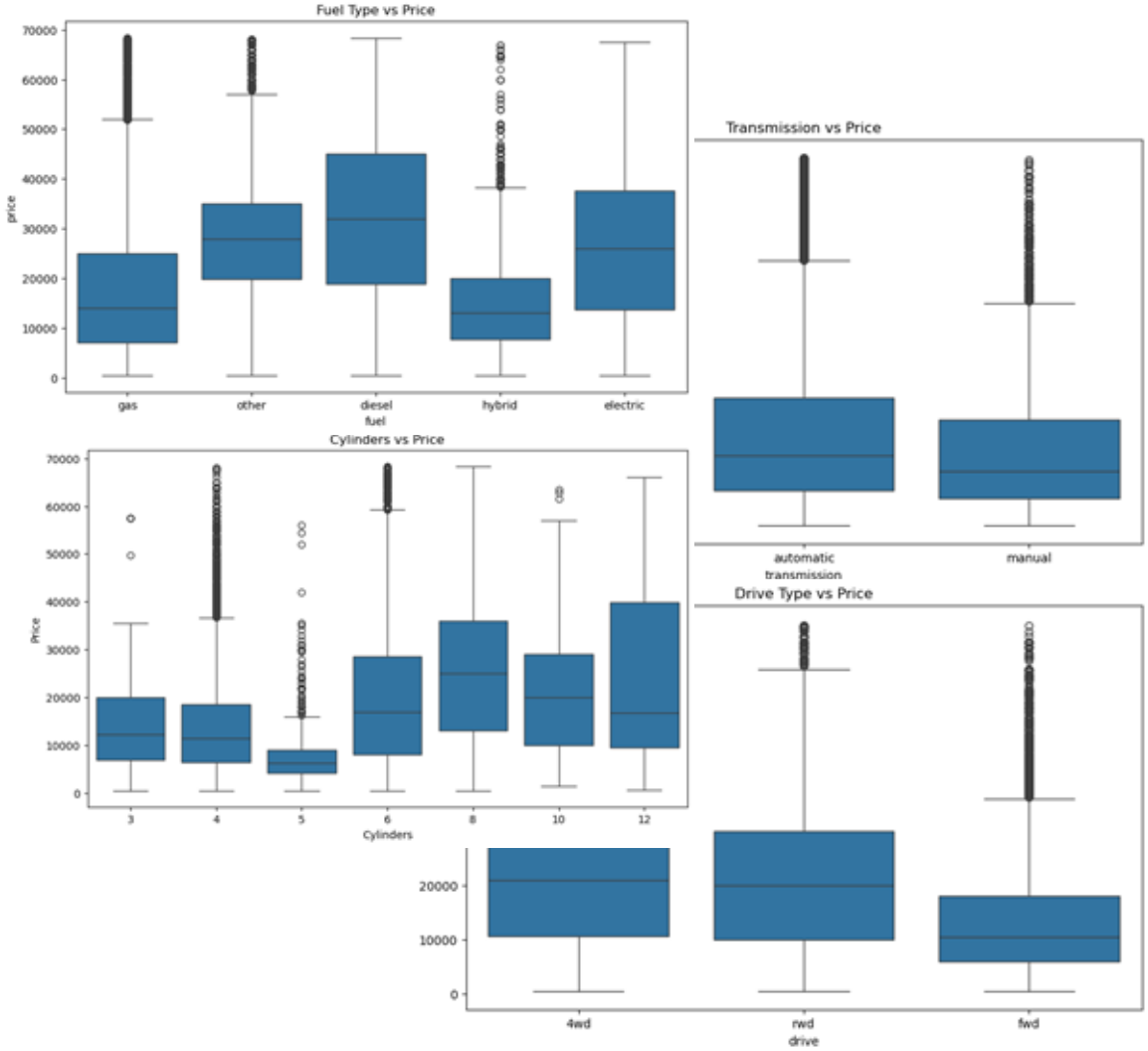
3.2 Categorical Variables and Price Analysis

Result Explanation:

The analysis of vehicle-related variables shows that transmission and drive align with expected price trends.

Diesel fuel vehicles have higher prices, which can be attributed to the dominance of pickup trucks and trucks in this category.

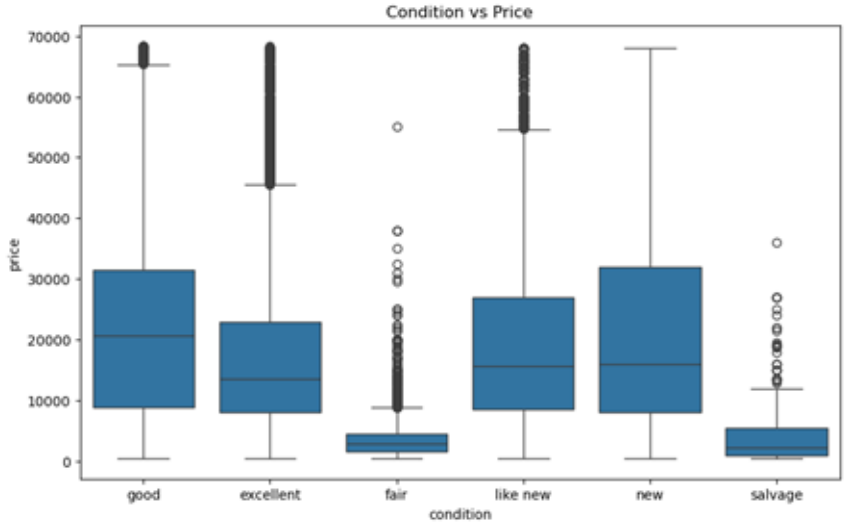
Additionally, 5-cylinder cars in the cylinders category are no longer in production, leading to their lower prices.

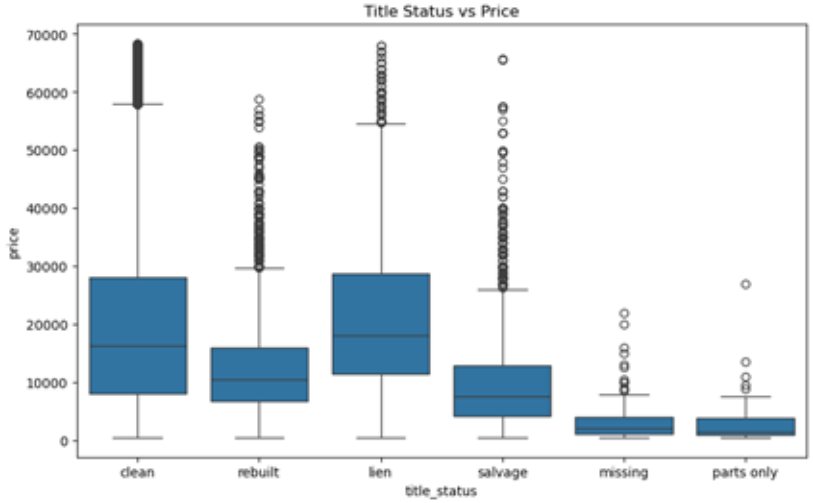


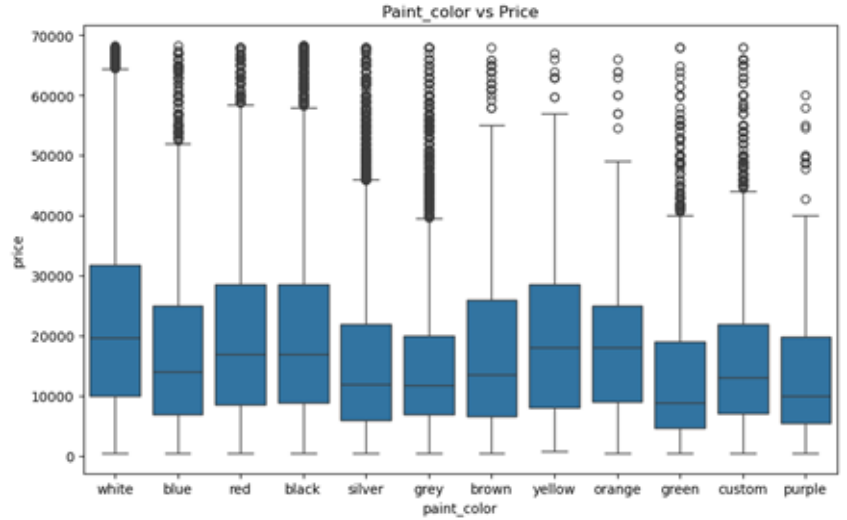
3.2 Categorical Variables and Price Analysis

Result Explanation:

condition, title_status, and paint_color don't show clear relationship with price. We can notice that the good condition car price and lien status car price are uncommon. And the cars with common color are a little bit higher in price.







3.3 Data Encoding and Standardization

Data Encoding:

To avoid introducing false meanings caused by direct labeling, we split the object variables into two types: high and low cardinality. The low cardinality variables can use label encoding and target encoding for the high cardinality variables.

```
region number of objects: 404
manufacturer number of objects: 41
condition number of objects: 6
fuel number of objects: 5
title_status number of objects: 6
transmission number of objects: 3
drive number of objects: 3
type number of objects: 14
paint_color number of objects: 12
state number of objects: 51
main_model number of objects: 301
```

Data Standardization:

To avoid disproportionately influence and biased predictions after target labeling, we applied standardization to ensure all features had a comparable scale.

#	Column	Non-Null Count	Dtype
0	num__odometer	274701 non-null	float64
1	num__region_encoded	274701 non-null	float64
2	num__manufacturer_encoded	274701 non-null	float64
3	num__state_encoded	274701 non-null	float64
4	num__main_model_encoded	274701 non-null	float64

4. Model Training and Evaluation

The goal of this section is to train the most suitable model for predicting car prices. To achieve this, we trained Ridge Regression, Decision Tree, Random Forest, XGBoost, and ANN, each selected for its potential to capture various patterns in the data and deliver reliable predictions.

4.1 Ridge Regression

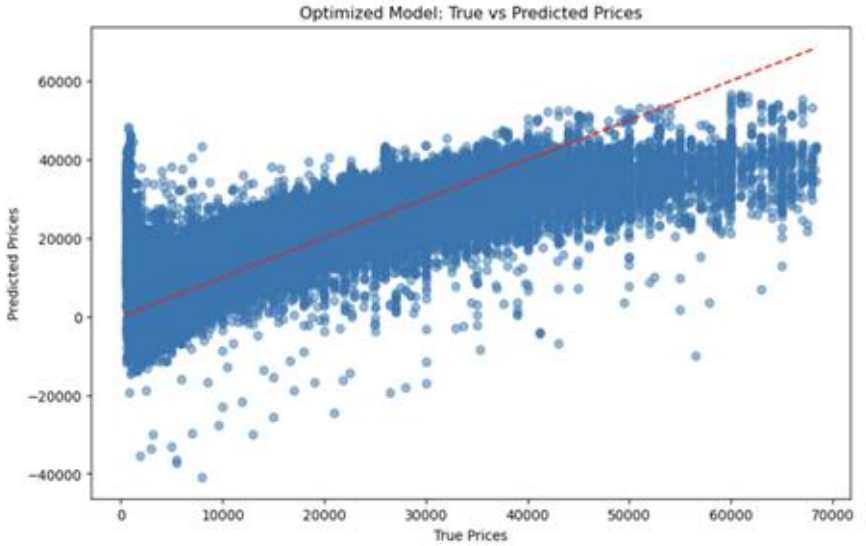
Model Introduction:

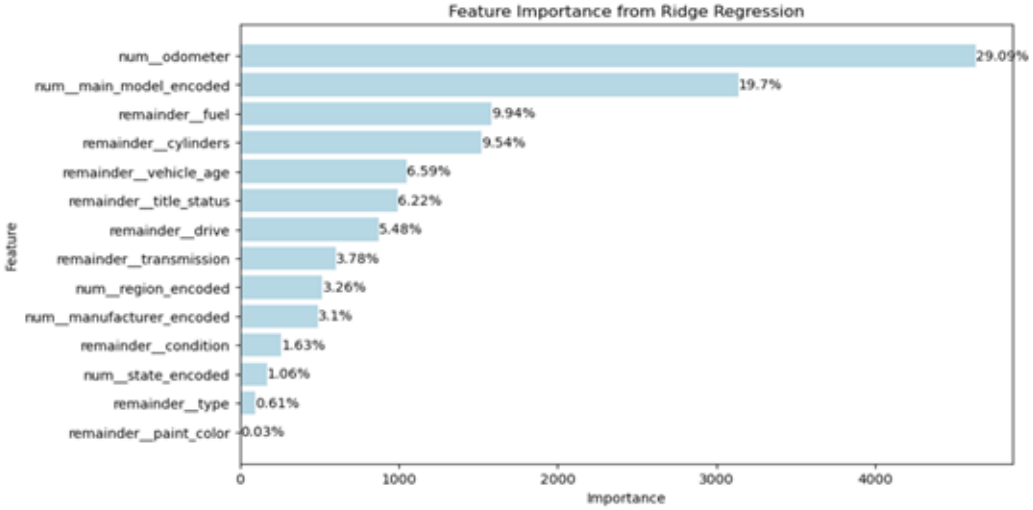
- Linear regression with L2 regularization
- Handles multicollinearity
- Reduces overfitting

Training Process:

- Trained with alpha=1.0 (default) to assess performance.
- Hyperparameter Tuning: GridSearch (α = 100)

Evaluation metrics	Initial model	Optimized model
R^2	0.7159	0.7159
MSE	49440513	49440397
RMSE	7031.39	7031.39





4.2 Decision Tree

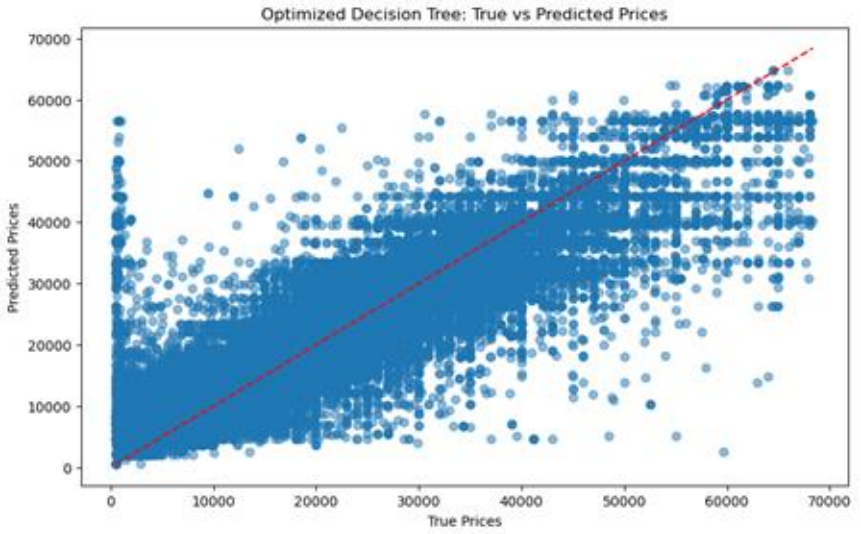
Model Introduction:

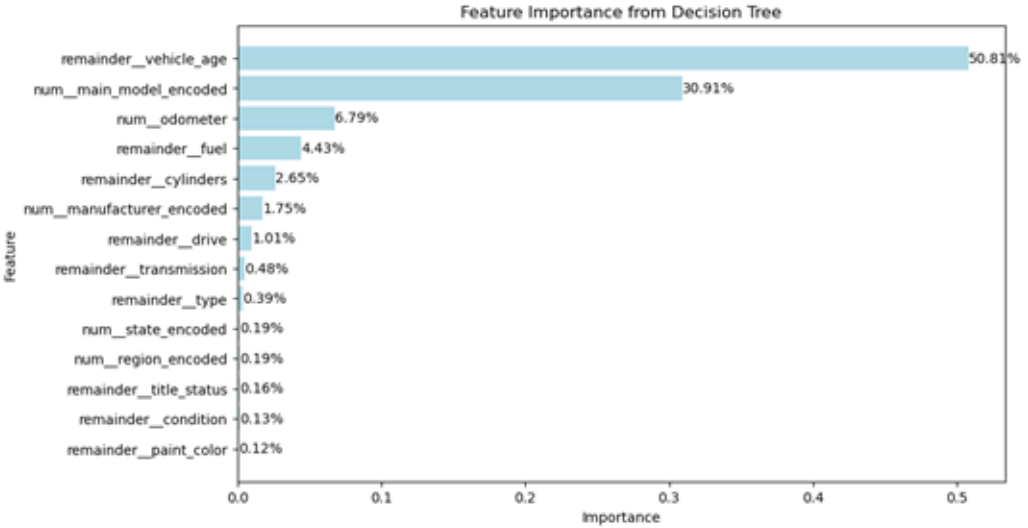
- Decision Tree is a hierarchical model that splits data based on feature conditions
- Handle Non-Linearity
- Baseline for Ensemble Methods

Training Process:

- Trained with max_depth=5
- Hyperparameter Tuning: GridSearchCV (best: max_depth=10, min_samples_leaf=5, min_samples_split=2)
- Cross-Validation confirming stable performance(R^2: 0.82, Std Dev: 0.0015)

Evaluation metrics	Initial model	Optimized model
R^2	0.7219	0.8255
MSE	48398042	30355975
RMSE	6956.87	5509.63





4.3 Random Forest

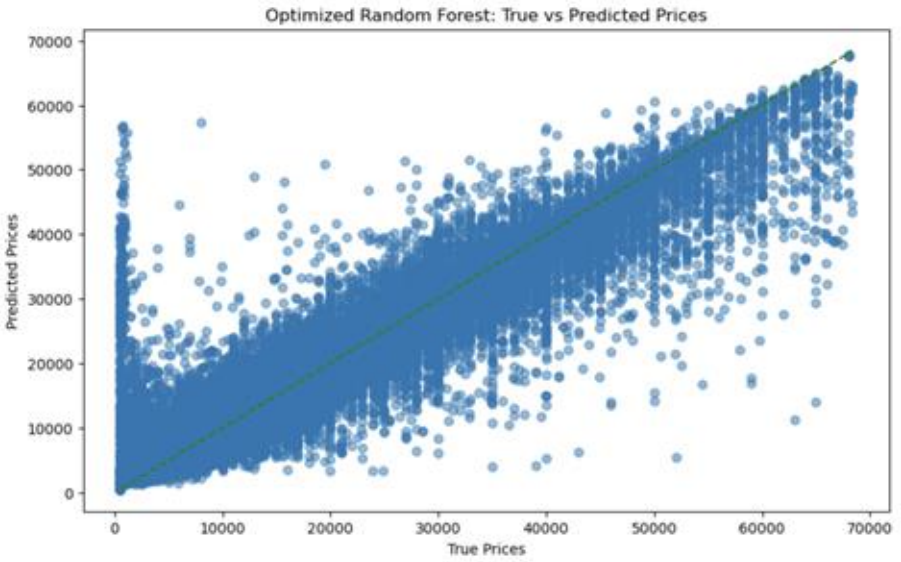
Model Introduction:

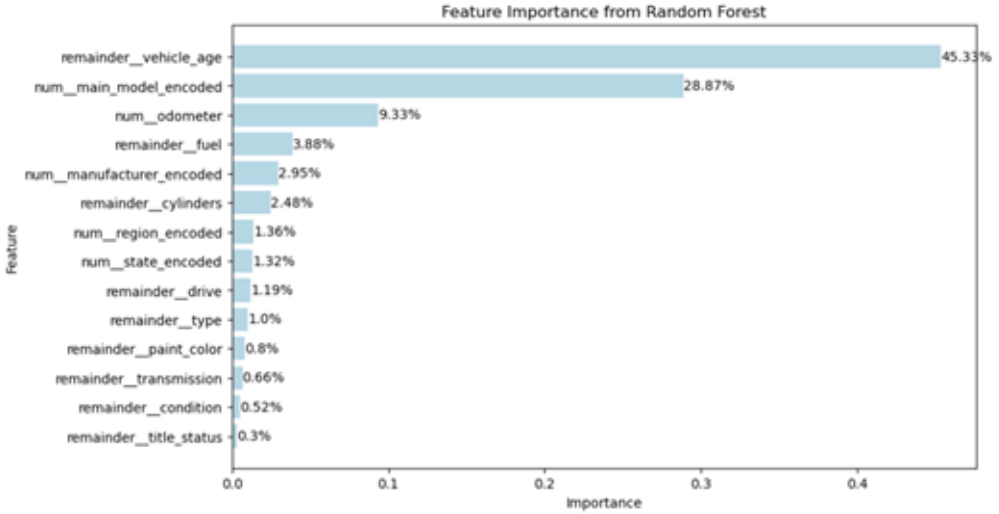
- Ensemble of decision trees for regression
- Handles non-linearity and interactions
- Reduces overfitting

Training Process:

- Adopted Decision Tree's best parameters as a baseline
- Bayesian Optimization
- Cross-Validation confirming stable performance(R^2:0.92, Std Dev:0.0036)

Evaluation metrics	Initial model	Optimized model
R^2	0.8427	0.9342
MSE	27378962	11444014
RMSE	5232.49	3383.90





4.4 XGBoost

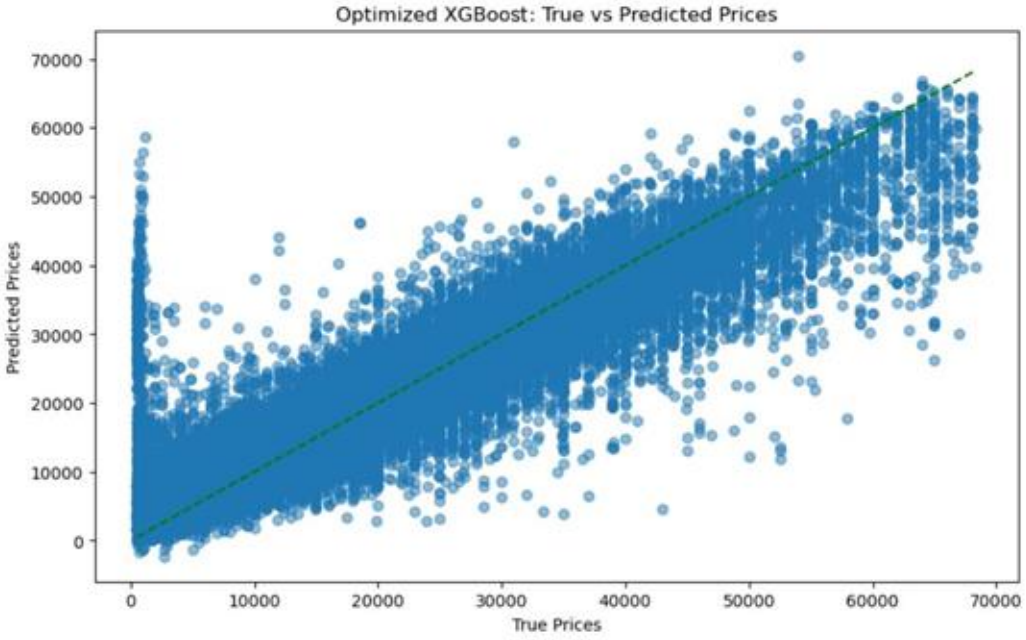
Model Introduction:

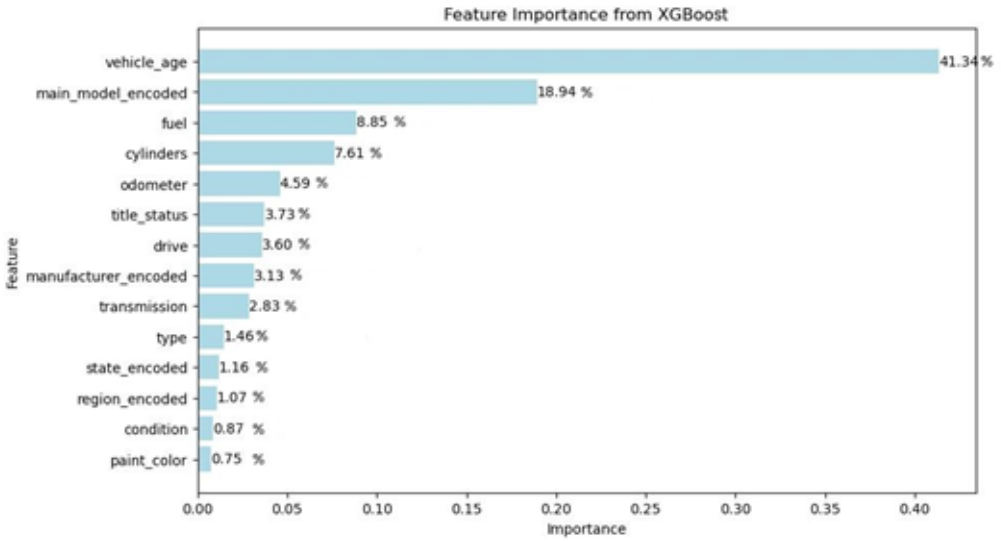
- Gradient Boosting framework
- Combines multiple weak learners to form a strong predictor
- Handles non-linearity, high-dimensional data
- Includes regularization

Training Process:

- Run default model
- Hyperparameter tuning: optuna
- Cross-Validation confirming stable performance

Evaluation metrics	Initial model	optimized model
R^2	0.8829	0.8932
MSE	20372885	18589705
RMSE	4513.63	4311.58





4.5 ANN

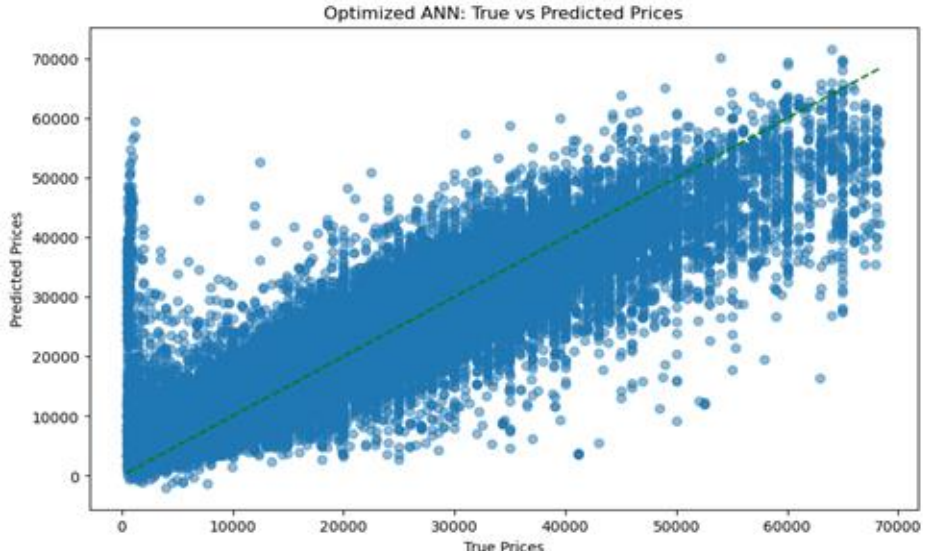
Model Introduction:

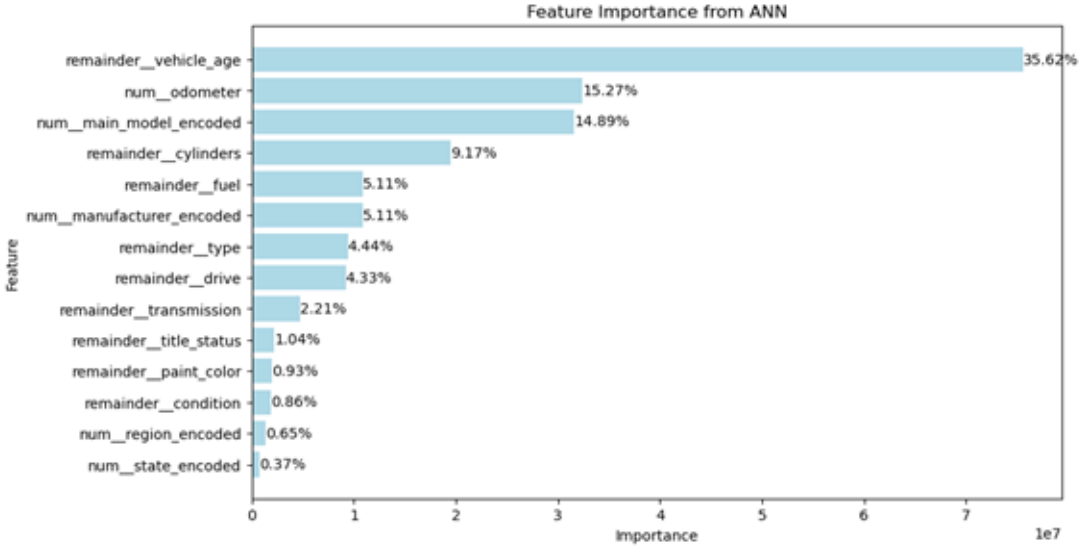
- Simulate the human brain, designed to recognize patterns and make predictions
- Highly flexible and adaptable to different types of data
- Can scale to very large datasets

Training Process:

- Adopt the Keras Sequential model with 2 layers
- Adjust hyperparameters using RandomizedSearchCV
- Cross-Validation confirming stable performance

Evaluation metrics	Initial model	Optimized model
R^2	0.8088	0.8407
MSE	33272596	27726860
RMSE	5768.24	5265.63





5. Conclusions and Further Development

Comparison:

	R^2	RMSE	Computational complexity	Merits	Demerits
Ridge Regression	0.7159	7031.39	Low	Handles multicollinearity	Struggles with nonlinear relationships
Decision Tree	0.8256	5509.63	Medium	Requires minimal preprocessing	High variance
Random Forest	0.9342	3382.90	High	Handles complex relationships	Higher cost
XGBoost	0.8932	4311.58	High	Prevent overfitting(regularization) Highly flexible	Higher cost Hard to interpret
ANN	0.8407	5265.63	High	Handles flexible, non-linear relationships	Hyperparameter sensitive Hard to interpret

5. Conclusions and Further Development

Feature importance interpretation(>5%):

Vehicle Age	Directly linked to depreciation Higher market prices expected for newer vehicles
Odometer	Directly linked to depreciation Lower mileage suggests a better condition
Main Model	Popular or premium models tend to maintain higher resale values
Fuel	Fuel-efficient vehicles or those with alternative energy sources (electric vehicles, hybrids) tend to have higher resale values
Cylinders	Vehicles with larger engines (more cylinders) typically offer higher power and performance

5. Conclusions and Further Development

Key insights:

- **Model performance metrics (based on R^2/RMSE):**

  Random Forest/ XGBoost

- **Efficiency:**

  Decision Tree/ Random Forest

- **Adaptability(based on Mean CV R^2 Score):**

  All models performed similarly.

Limitations:

- Interpretability
- Sensitivity to Hyperparameters
- Scalability
- Computational Complexity

6. Recommendations

Potential Application Scenarios:

- **Users (Seller/Buyer)**

  Price guidance 📊/ Better decision-making ✅
- **Commercial Platforms**

  Dynamic pricing models ⚙️/ Fraud detection 🔍
- **Insurance Pricing**

  Risk assessment 📉/ Premium calculation 💰
- **Academic Research**

  Market trend analysis 📈/Predictive modeling 🧠

Thank you!

